

Supplementary Material

A Data Availability

We provide all the inference code and results in the zip file during the revision process. Once published, we will make all data and code available online.

B Prompts

B.1 Prompts for VLMs

Here we present the final prompts used for prompting the VLMs. System prompts for estimation approach:

“You are attending a numerical perception test, perform your best. Only respond with a single number for your count result and nothing else.”

System prompts for localizing approach:

“Return bounding boxes as an array with labels. Never return masks. The bounding boxes should be in `Ymin, Xmin, Ymax, Xmax` format, with coordinates normalized from 0 to 1000. If an object appears multiple times, assign each instance a unique label based on its distinct characteristics (e.g., color, size, position, etc.).”

While for the labeling approach, the system message was:

“You are attending a counting ability test, perform your best counting ability. You are going to count certain objects given an image. Count from left to right, top to bottom. Assign each object a unique label. Return the labels as an array of strings and an integer indicating how many objects there are in total”

Then, for each image, we paired it with a user message specifying the object category:

“Output the positions of all the `[category name]` in the image.”

for the localizing approach, and for the labeling approach:

“How many `[category name]` are there in the image?”

The prompt for the direct numerical estimation approach was instead defined by the following system message:

“You are attending a test on numerical perception. Perform your best numerical ability and respond to the question with just one number.”

and the user message:

“How many `[category name]` are there in the image?”

B.2 Prompts Explored

We also explored different system prompts as well as user prompts to maximize the models’ performance under each approach. We use separate validation sets for all datasets to decide which prompt results in the best performance.

Label and count As for the label approach, we mainly focus on the system messages:

“You are attending a test for numerical perception and visual reasoning. You need to count how many objects there are in the image. Give each counted object a unique label and return the final count.”

“Please perform your best counting ability. Count from left to right, top to bottom. Assign each counted items a label and report the final count.”

Point, label and count We also tried "Detect all the [category name] in the image." for user message and "Detect the all of the prominent items in the image. The box_2d should be [ymin, xmin, ymax, xmax] normalized to 0-1000." for system message.

C Threshold Optimization

For PseCo, we used separate validation sets for all datasets by applying a grid search from 0.01 to 0.99 with an increment of 0.01 to find the best threshold on the metrics. The optimal thresholds for PseCo were determined as follows: for FSC-147, a threshold of 0.48 achieved an MAE of 3.7 and an Acc. of 0.28; for FSCD-LVIS, a threshold of 0.68 resulted in an MAE of 14.06 and an Acc. of 0.03; and for SolidCount, a threshold of 0.60 yielded an MAE of 6.28 and an Acc. of 0.09. These optimal thresholds were obtained using a separate, validation dataset, which is not always available in real-world scenarios. Thus, we report the performance of PseCo without those fine-tunings in the main text.

D Challenging Examples of FSCD-LVIS

In this section, we discuss a set of practically intractable counting cases from the FSCD-LVIS dataset as shown in Fig. S1. These samples encompass various sources of complexity: (1) heavy occlusion and object clustering, such as in the Toy and Fruit categories, where objects are densely packed and partially overlap, making individual instances hard to distinguish; (2) low object salience, where targets are visually inconspicuous due to color similarity with the background (e.g., Drink Container) or low contrast conditions (e.g., Necktie in a grayscale group photo); (3) truncated or partially visible objects at image boundaries, which introduce ambiguity in determining the presence or completeness of targets (e.g., Cow, Backpack); and (4) ambiguous category definitions or uncertain



Fig. S1. Examples of practically intractable counting cases from FSCD-LVIS dataset. On top of each image is the target object’s category name. These images exhibit extreme occlusion, low object visibility, and ambiguous object boundaries, making even human annotation unreliable and consistent counting nearly infeasible.

countability, such as Log, where the boundaries of what constitutes a single instance are unclear. Due to the near-impossibility of obtaining consistent counts from these examples, we argue that such samples are unsuitable for reliable evaluation. We therefore recommend shifting focus toward the construction of more controllable and clearly defined datasets to facilitate meaningful progress in object counting research.

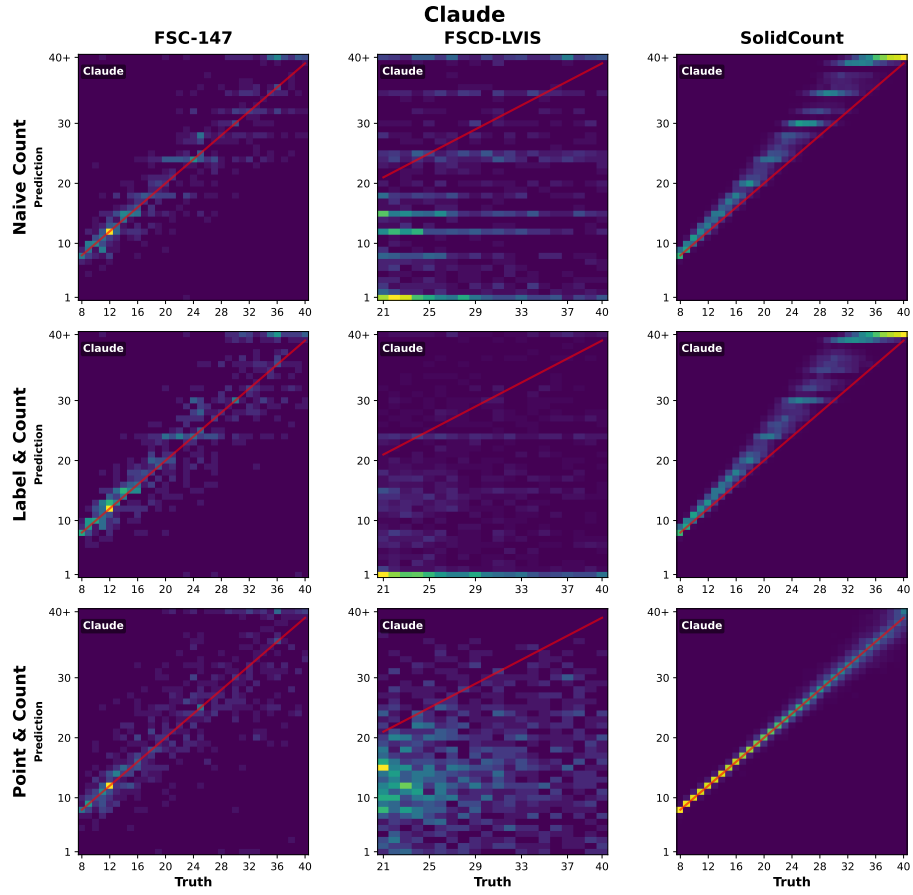


Fig. S2. Confusion matrices across all datasets. The x-axis represents the ground truth, while the y-axis denotes the predictions, with the final row ("40+") aggregating all predictions where $n > 40$. The red line indicates the perfect responses.

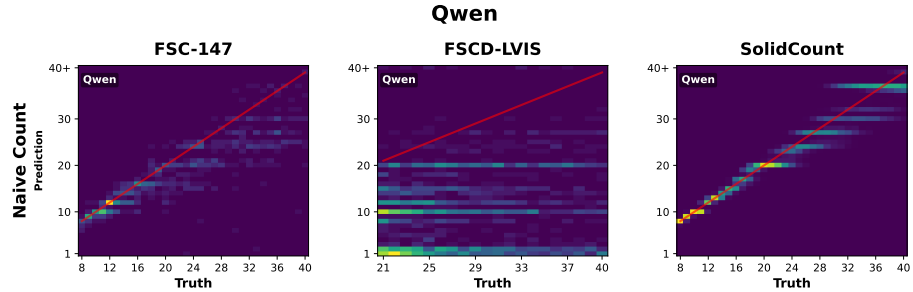


Fig. S3. Confusion matrices across all datasets. The x-axis represents the ground truth, while the y-axis denotes the predictions, with the final row ("40+") aggregating all predictions where $n > 40$. The red line indicates the perfect responses.

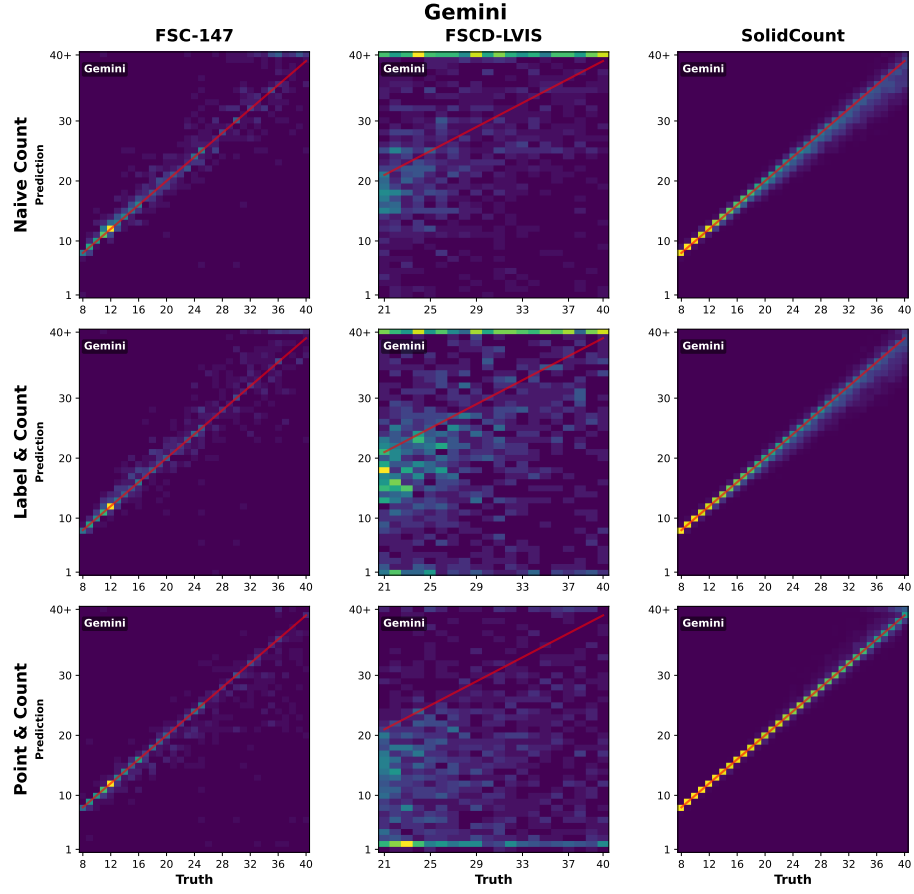


Fig. S4. Confusion matrices across all datasets. The x-axis represents the ground truth, while the y-axis denotes the predictions, with the final row ("40+") aggregating all predictions where $n > 40$. The red line indicates the perfect responses.

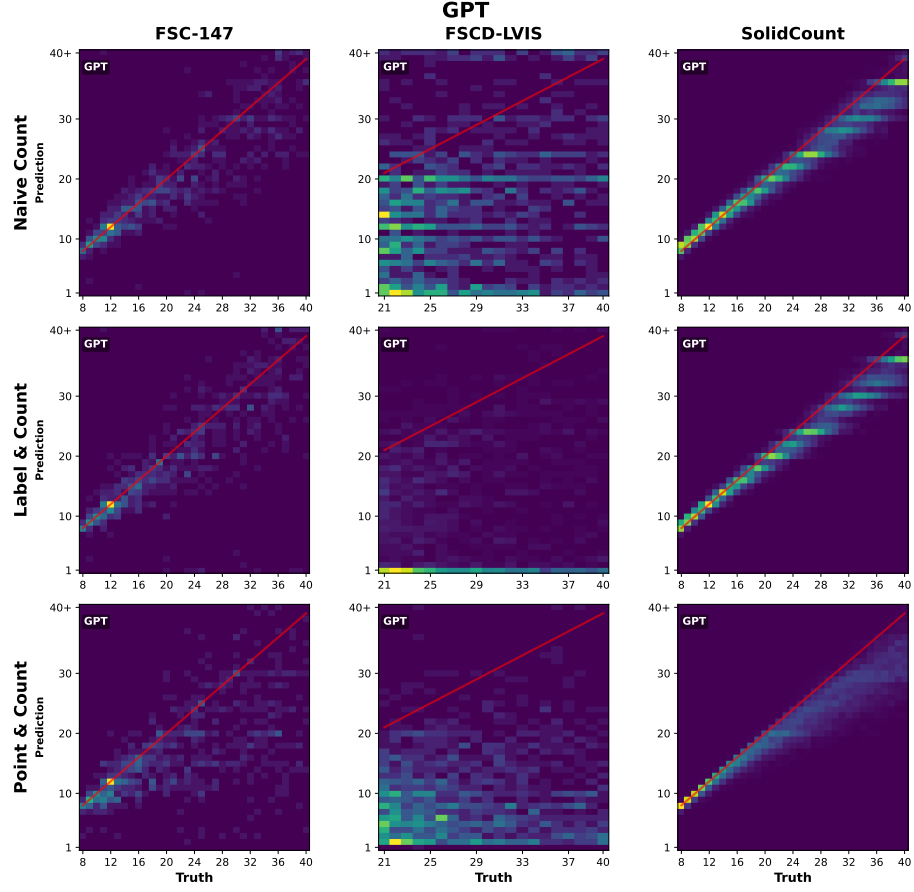


Fig. S5. Confusion matrices across all datasets. The x-axis represents the ground truth, while the y-axis denotes the predictions, with the final row ("40+") aggregating all predictions where $n > 40$. The red line indicates the perfect responses.

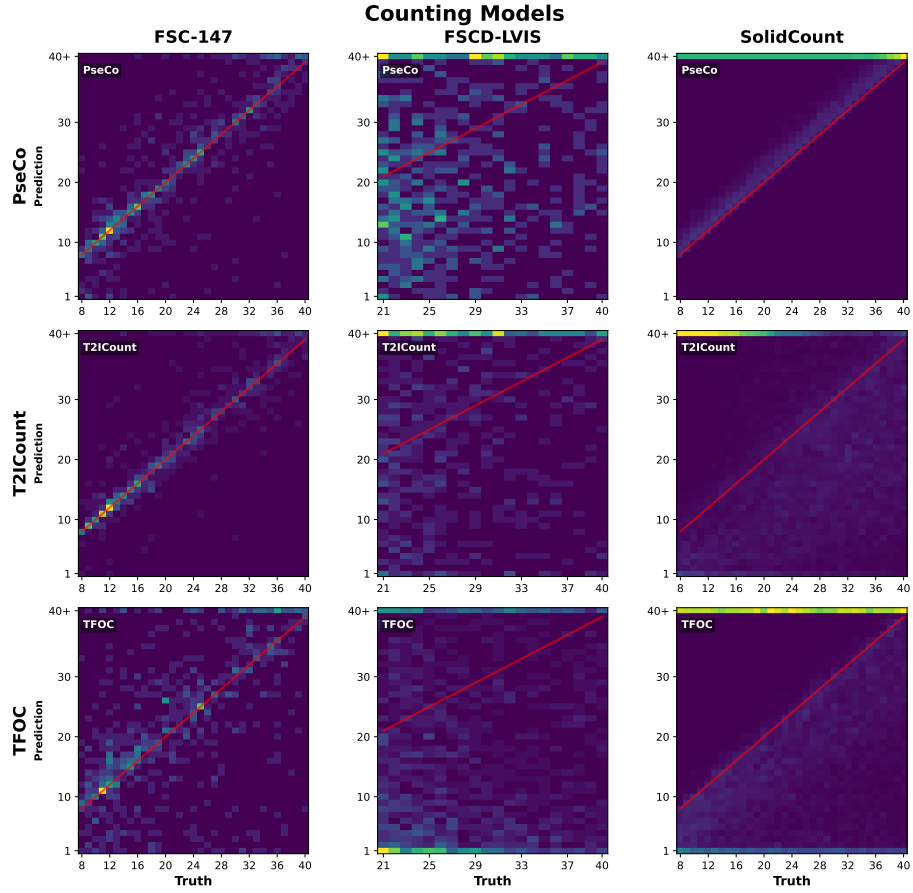


Fig. S6. Confusion matrices across all datasets on **Counting Models**. The x-axis represents the ground truth, while the y-axis denotes the predictions, with the final row ("40+") aggregating all predictions where $n > 40$. The red line indicates the perfect responses.

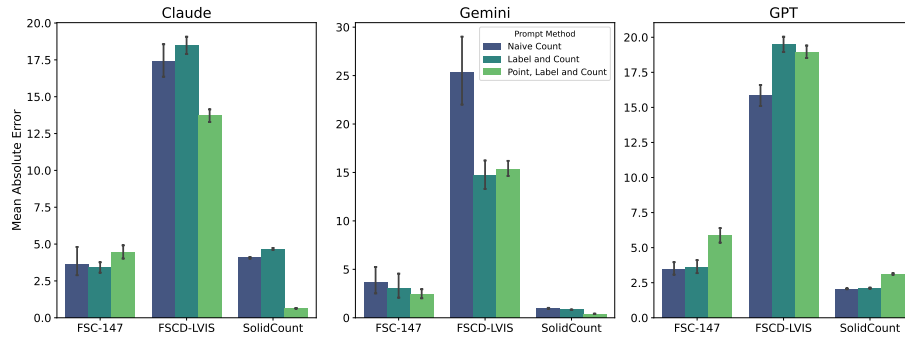


Fig.S7. Bar chart displaying the MAE of three prompting strategies across all datasets. The height of each bar represents the mean MAE, and the error bars indicate the 95% confidence interval of the mean.